Contents

1 Introduction **1**
1.1 Aim . . . 1
1.2 Sources . . . 2

2 Theory **2**
2.1 Interaction of γ rays with matter . . . 3
2.1.1 Photoelectric effect: . . . 4
2.1.2 Compton scattering . . . 5
2.1.3 Pair production . . . 5
2.2 Scintillator . . . 6
2.3 Photomultiplier tube (PMT) . . . 6
2.4 Multi channel analyzer . . . 6

3 Data **6**
3.1 Raw Data . . . 6
3.1.1 Spectrum data for Co^{60} . . . 6
3.1.2 Spectrum Data for Cs^{137} . . . 8
3.2 Analysing the data . . . 8
3.2.1 Graphs . . . 8
3.2.2 Calibration of MCA and finding photopeak energy of Cs^{137} . . . 11
3.2.3 Photopeak Efficiency of Co^{60} peaks . . . 11
3.2.4 Variation of resolution with PMT voltage . . . 12
3.2.5 Variation of calibration constnat (α) with applied PMT voltage . . . 13
3.2.6 Analysis of Co^{60} spectrum . . . 14

4 Discussion **15**

5 Conclusion **15**

1 Introduction

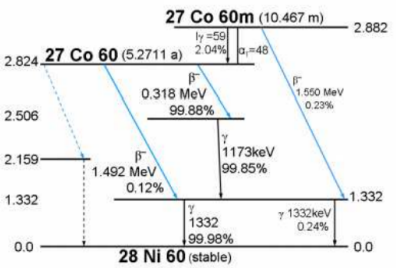
1.1 Aim

To detect and study γ radiation produced by Cs^{137} Co^{60} with a scintilator counter using a multi channel analyzer.

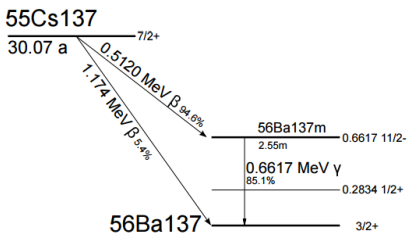
- Given that energy of gamma rays in decay of Co^{60} to be 1.17 MeV and 1.33 MeV. We find chanel number corrsponding to photopeack of these two enrgy to find relation between chanel number and energy.
- We find energy of gamma ray in decay of Cs^{137}
- We find photopeak efficiency of Co^{60}
- We also study variation of resoltion for two Co^{60} peaks with variation of PMT voltage.

1.2 Sources

γ-ray source used in this experiment are Cs^{137} (one peak) and Co^{60}(two peaks)



(a) Decay Scheme of Co^{60}



(b) Decay Scheme of Cs^{137}

2 Theory

In this experiment we study about detection of gamma(γ) by scintillator detector. It is a multi-step process: the gamma ray enters a NaI:Tl scintillator crystal where it produces a rapidly moving free electron that, in turn, loses its energy by excitation of the ions in its path as it travels through the crystal. This excitation energy is given off in various ways, one being emission of visible light (fluorescence). Thus a single high energy gamma ray entering the scintillator produces a flash of low energy photons. These photons are directed to the photosensitive surface of a photomultiplier tube(PMT), where they eject electrons via the photoelectric effect. The electrons are collected in the photomultiplier and amplified to yield a current pulse, which is converted to a voltage pulse whose height is proportional to the number of photoelectrons and is thus proportional to the number of photons reaching the tube, which in turn is proportional to the initial energy of the fast electron.

Thus the amplitude of each pulse is related to the energy of the electron freed by the gamma ray.The energy signal from the photo multiplier is usually quite weak; therefore it must be amplified before analyzing it. The output pulses of the PMT are amplified and then recorded. Therefore the output pulse is proportional to the energy deposited in the scintillator. The proportionality constant has to be determined experimentally. This process is called calibrating the detection system.

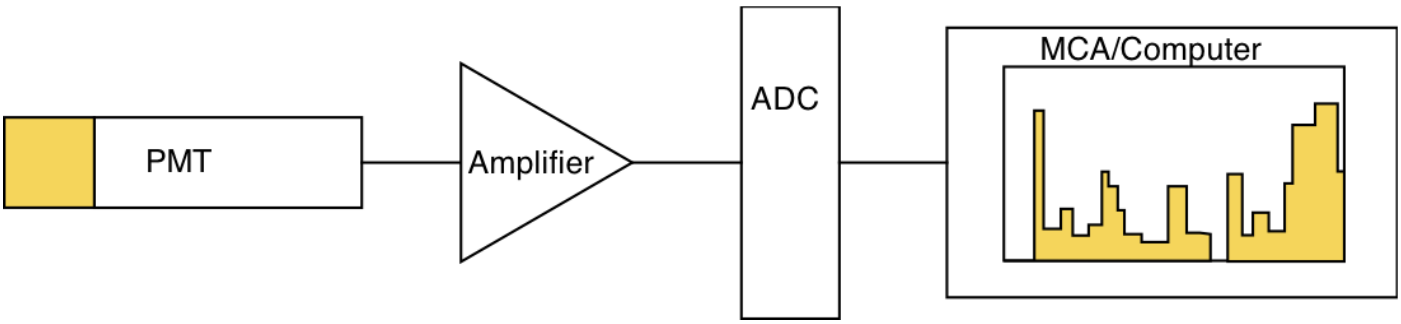


Figure 1: Schematic of the data acquisition with PC

The amplified pulses are analyzed by an Analog to Digital Converter (ADC) which determines the pulse height and converts it into a number between 0 and 1023. "Number of times a particular ADC value is observed" is counted. So We have a total of 1024 counters. Each ADC value is called a channel and the counter value is called the channel contents. If we take data for a while we will see that some

values occur more frequently than others. The total set of counter values and the corresponding ADC values is called a spectrum. When we plot the graph x-axis corresponds to the ADC value and the y-axis represents the corresponding counter values or channel contents. The process described above is performed by a Multi Channel Analyzer (MCA). Voltage pulses are studied using either a single- and multi-channel analyzer. A single channel analyzer (SCA) counts the number of voltage pulses whose height falls within a given (adjustable) window of values, while a multichannel analyzer (MCA) sorts the pulses according to height and counts the number in each window to give a spectral (energy) distribution of the fast electrons.

Figure-2 shows a typical MCA spectrum. In order to relate this spectrum to the nuclear decay, we need to understand how gamma rays interact with matter.

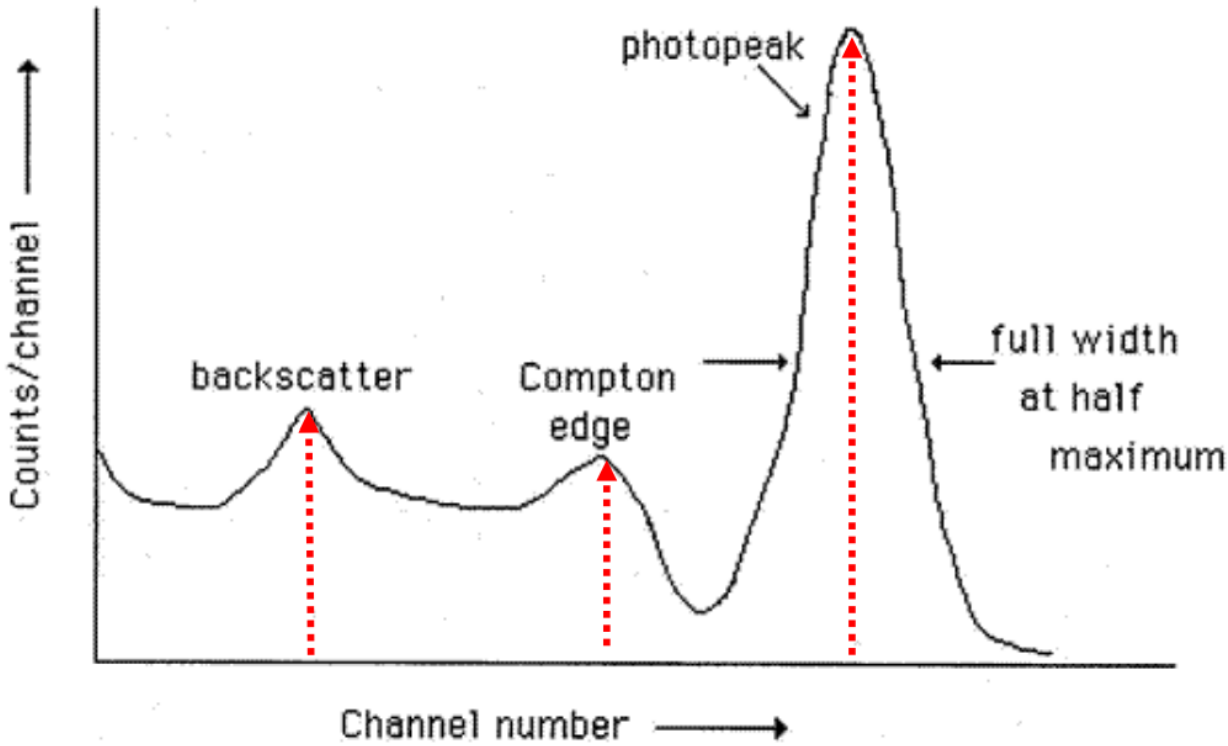


Figure 2: NaI:Tl Spectrum for Cs^{137}

2.1 Interaction of γ rays with matter

Gamma rays are high-energy photons. Photons have no mass and no electrical charge, therefore they cannot directly ionize matter, neither gamma rays. Despite this fact, gamma rays ionize matter via indirect ionization. Although a large number of possible interactions are known, three key interaction mechanisms with matter are as following.

- Photoelectric effect
- Compton effects
- Pair production

Which of these processes contributes the most is mainly dependent on theatomic number (Z) of the material and the energy (E) of the photon. Figure-3 shows that At low energies and with high Z materials the photoelectric effect is main interaction process. At intermediate energies and in low Z

materials, the Compton scattering is dominating. At very high energies pair production is the most dominant interaction process.

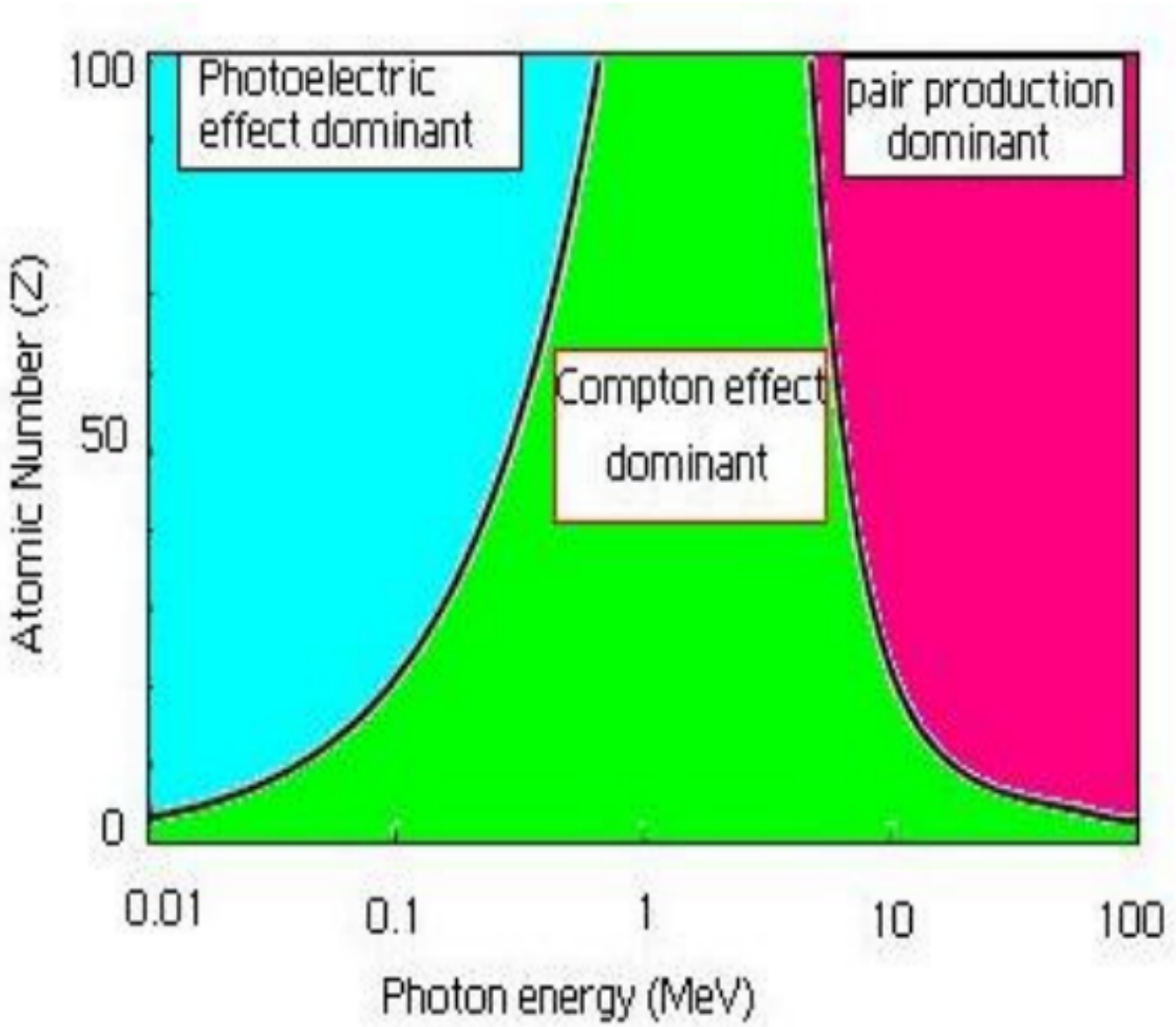


Figure 3: Z-E diagram

2.1.1 Photoelectric effect:

When the photon collides with an atom, it may impinge upon an orbital electron and transfer all of its energy to this ejecting it from the atom. When the incident photon energy $h\nu$ exceeds the electronic binding energy (or ionization energy) E_B, the electron is ejected with a kinetic energy

$$E_{kin} = h\nu - E_B \tag{1}$$

This process, in the course of which the photon disappears completely, takes place exclusively in the direct vicinity of the nucleus because the impulse the photon has due to its energy and velocity can be transferred to an electron for a small part only as mass of the electron is low. The rest of the impulse must therefore be transferred to the nucleus. So, the process only takes place with K or L-electrons and occurs more often with substances with a high atomic number (Z). When the photon energy is too high, a nucleus with a high Z cannot handle the surplus of impulse either,that is why the photoelectriceffect only occurs up to a limited energy value.

If for example an atomic K-electron is ejected as a result of the photoelectric process Very shortly after, another bound atomic electron will "fall" into the K shell vacancy (or cascade down sequentially) with the subsequent emission of X-rays. These X-rays will have a large probability of producing light pulses in the scintillator by exciting other loosely bound electrons. These processes (initial photoelectron ejection and subsequent X-ray production and interaction with the crystal) will usually happen within the resolution time of the counter, so that these successive light pulses add and can not be distinguished from one another. Therefore in the end the photomultiplier output pulse will correspond to the full gamma ray energy, if the photoelectron stops in the crystal and if no light escapes the crystal. **Thus the photoelectric effect results in a peak, called the photopeak, at an energy equal to that of the incoming gamma ray**.

2.1.2 Compton scattering

In this case the gamma ray is not absorbed, but rather scattered through an angle θ by an electron, which recoils and carries away some of the gamma rays energy E. The scattered electron energy is then detected in the photomultiplier. The scattered gamma ray escapes from the scintillator because the probability that a gamma ray Compton scatters in a typical size scintillator is quite small (0.01 to 0.1), which means we are unlikely to detect a gamma ray that has undergone two Compton scatterings.The energy of the Compton-scattered gamma ray $E_\gamma^{'}$ as a function of the scattering angle θ and the initial energy E_γ is given by

$$E_\gamma^{'} = \frac{E_\gamma}{1 + \frac{E_\gamma}{mc^2}(1 - cos\theta)} \tag{2}$$

Where c is the speed of the light and m is the mass of the electron. From equation(2) we see that the energy of the scattered electron, which is the energy loss of the gamma ray, will vary from zero (when $\theta = 0^\circ$) to a maximum of $\frac{2E_\gamma^2}{2E_\gamma + mc^2}$ (when $\theta = 180^\circ$). This maximum energy is called the **Compton edge.** The energy distribution of Compton scattered electrons is essentially a constant. So the Compton spectrum produced by a photomultiplier tube is an almost flat plateau from zero energy up to the **Compton edge** where it drops off sharply (at a rate limited by the energy resolution of the tube).

The discussion above refers to gamma rays that are Compton scattered by electrons within the scintillator. It is also possible for a gamma ray to be Compton scattered into the scintillator from an interaction outside the scintillator. In this case the observed signal is from the scattered gamma and not from the recoiling electron. The scattered gamma ray could then be detected through the photoelectric effect. However, because of the geometry of the detector, most of the gamma rays scattered into the scintillator will have been scattered through a large value of θ (almost 180°). But cosθ varies only slowly with θ for θ near 180°, which means [see equation(2)] that these gamma rays will have energies close to $\frac{mc^2 E_\gamma}{2E_\gamma + mc^2}$ resulting energy peak is called the **backscatter peak**. It can be enhanced by placinga sheet of lead around the outside of the scintillator.

2.1.3 Pair production

If the incoming gamma ray energy is above $1.02MeV = 2mc^2$,the rest mass of an electron-positron pair, the gamma ray can spontaneously create an electron-positron pair and be totally absorbed. If both the electron and positron lose all of their kinetic energy while still in the scintillator, they would produce a photomultiplier pulse corresponding to an energy $2mc^2$ below the gamma ray energy $(E - 2mc^2)$.

2.2 Scintillator

In this experiment, scintillator detector with NaI:Tl crystal is used to detect gamma ray. As described in previous section, that photoelectric effect will dominate, which depends on the atomic number of the interacting matter. Thus NaI is grown with some heavy nuclei (e.g. Thalium;Tl). The resolution of such detector is about 10%. High purity Germanium may be used as detector to obtain greater resolution, but is also highly expensive. Depending on the size of the NaI crystal, occasional escape of electrons, X-rays or γ rays.

2.3 Photomultiplier tube (PMT)

After generation of secondary photons, they hit the photo-cathode of PMT, releasing electrons by photoelectric effect. These electrons are then focused to a series of Dynodes, each of which are at positive potential than the previous (can be achieved by a voltage divider). From each dynode, by positive potential, electrons are accelerated, which striking the next dynode, releases more electrons. In this way, up to the anode, a large number of electrons are produced, which leads to a current, which is then converted into a voltage pulse. The height of the voltage pulse is proportional to the initial energy of the photons, hence accounts for the energy of the γ ray. The efficiency of a PMT tube is about 30%

2.4 Multi channel analyzer

The MCA counts number of counts for all the relavant energy range at the same time. As point out in theory section. Energy of the detected signal is praportional to chanel number. The praportionality constant is determined experimentally. Since we know two energy value E_A and E_B for Co^{60} we can find corresponding chanell number A and B to find this praportinality constant α, say

$$\alpha = \frac{E_A - E_B}{A - B} \tag{3}$$

3 Data

A known source of $Cs^{137} and Co^{60}$ are choosen. We turn on the detectors. We collect data using a computer interface. Various parameters like gain, PMT voltage etc are setted using computer interface. Parameters value are setted as following:
Gain: 8
LLD(Lower Level Discriminator):1
Time for a single reading : 600 Seconds. There is some dead time during which detector can not record spectrum data. So each data is taken for actaully $600 + dead\ time$ seconds.

3.1 Raw Data

After setting paramters as specified above we take spectrum data for Co^{60} and Cs^{137}.

3.1.1 Spectrum data for Co^{60}

We aim to to study variation of resolution of the detector at Co^{60} photopeaks with applied PMT voltage. So we take spectrum data at PMT votage 500V, 620V,620V,630V,...,720V. Spectrum data at different PMT voltage is presented in table-1.

Chan.	Counts at different PMT Voltage(First row represnt PMT voltage other rows contains count data)											
No.	500	620	630	640	650	660	670	680	690	700	710	720
0	0	0	0	0	0	0	0	0	0	0	0	0
1	0	0	0	0	0	0	0	0	0	0	0	0
2	0	0	0	0	0	0	0	0	0	0	0	0
3	0	0	0	0	0	0	0	0	0	0	0	0
4	22732	11657	15611	25431	24142	43140	47844	83162	108138	135982	162077	19190
5	29726	15388	21952	38648	30067	58561	72576	129911	178197	235300	289303	36659
6	31606	15721	20143	33984	32648	63470	73150	147290	207868	275487	340836	44368
7	34390	14883	18468	30810	28121	59696	73632	161127	241974	338521	431299	57594
8	37525	14442	15805	22213	19731	38595	48723	113869	183675	279733	381677	55210
9	39779	14538	14373	16113	15248	23347	26945	61798	101399	162798	234927	36851
10	42175	14631	14395	14442	13825	17893	19315	39184	61634	98130	144562	23612
11	44740	15106	14364	14204	13663	16109	16463	29174	44087	68758	99982	16748
12	48328	15328	14496	14028	13650	15110	14955	24179	34841	52220	75353	12671
13	52259	16136	14714	14236	13575	14914	14447	21725	29188	42533	59636	10018
14	56109	16574	14852	14191	13626	14822	14339	20496	26411	36226	49333	80875
15	59423	17834	15405	14509	13863	14931	14486	20036	24798	32638	43401	68342
16	60920	19161	16520	14752	14290	15586	14963	20823	26020	32832	42187	64167
17	60299	20539	18026	16476	16084	19409	19349	29348	36416	45135	55130	75803
18	58624	20707	18391	15052	14851	15567	13875	20855	24534	30065	37473	53369
19	55736	20275	19205	15320	14217	14481	13265	17927	21445	26683	32580	46268
20	53199	18981	19559	16446	15048	14704	13094	18153	22184	27534	33011	45120
21	51486	17791	19272	17542	16004	15499	14085	19457	24188	30377	37197	50444
22	49422	16335	18246	18660	16829	16665	14980	21595	27693	35461	44248	60417
23	47724	15470	16690	19019	17807	17847	15471	23200	29977	38900	48442	66584
24	46391	14582	15385	17702	17517	17518	14112	21034	26595	34101	43093	58599
25	44524	14248	14219	16240	16697	16581	12608	16110	17979	21947	26455	35686
26	44530	14135	13667	15023	15342	15504	12283	12093	11745	12695	14724	19436
27	43643	14322	13369	13988	14575	15326	12847	11219	9881	9854	10638	12671
28	42843	14285	13291	13227	13941	14726	13582	11642	10004	9280	9339	10346
29	42731	14475	13213	12541	12811	14022	13892	12410	10089	8957	8726	9069
30	42393	14459	13098	12173	12235	13379	13582	13027	10544	8783	8427	8390
31	42200	14651	13057	11765	11566	12348	12754	13467	11188	9225	8259	7811
32	42371	14590	13411	11608	11533	12072	12086	13481	11864	9486	8164	7493
33	41778	14710	13168	11843	11091	11166	11365	12904	12349	10018	8188	7449
34	42420	14886	13507	11719	11078	10856	10696	12331	12458	10356	8485	7391
35	42394	14873	13445	11584	11142	10332	10014	11709	12421	11139	9007	7491
36	42105	14755	13636	11824	11177	10572	9395	11169	12060	11505	9340	7323
37	41692	14938	13499	11974	11390	10355	9156	10669	11446	11963	9890	7409
38	42104	14964	13785	11999	11389	10254	8812	10103	10851	11626	10445	7590
39	41662	14896	13659	11841	11432	10169	8982	9704	10392	11196	10833	7895
40	41611	14766	13463	12046	11248	10422	8506	9394	10063	10738	10969	8231
41	41393	15083	13649	11914	11439	10298	8652	9361	9510	10315	10986	8610
42	41803	15313	13826	12231	11449	10233	8633	9118	8983	9703	10372	9137
43	41649	15621	13871	11849	11411	10183	8535	8894	8748	9469	9997	9686

44	42355	15741	13703	12365	11599	10557	8608	8823	8628	8862	9748	9820
45	42647	15756	13986	11966	11674	10527	8647	8845	8416	8642	9207	9818
46	43192	16352	13973	12236	11585	10607	8624	8774	8477	8300	9002	9663
47	43388	17327	14207	12023	11569	10677	8763	8887	8315	8036	8541	9624
48	44077	18103	14559	12079	11694	10610	8740	8856	8364	8210	8033	9125
49	44318	19240	14623	12110	11712	10575	8628	8904	8189	7797	7797	8871

Table 1: Co^{60} Spectrum Data from MCA

The full data can be found at this link: https://goo.gl/UozgMV

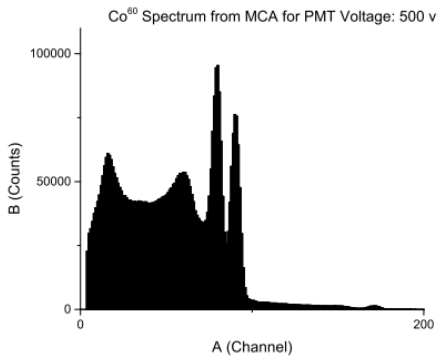
3.1.2 Spectrum Data for Cs^{137}

Spectrum data for Cs^{137} cab be found at this link: https://goo.gl/gYDMJT

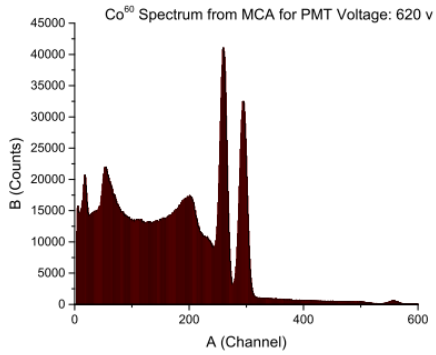
3.2 Analysing the data

3.2.1 Graphs

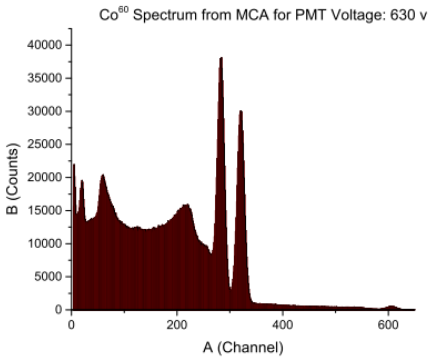
To visualize the energy spectrum of Cs^{137} and Co^{60} and also how this energy spectrum for Co^{60} changes with variation of applied PMT voltage we have plotted the histogram of the data in table-1 of counts vs chanel number for Co^{60} at a range of PMT voltages and for Cs^{137} at $650v$. These histograms are presented in figure-4.



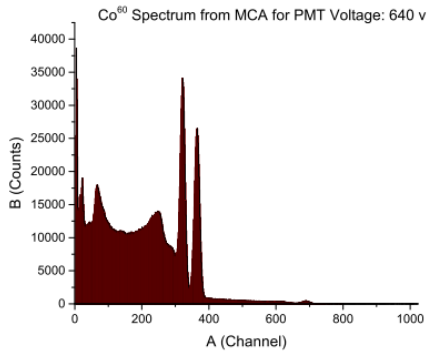
(a) Co^{60} spectrum for PMT voltage: 500 v



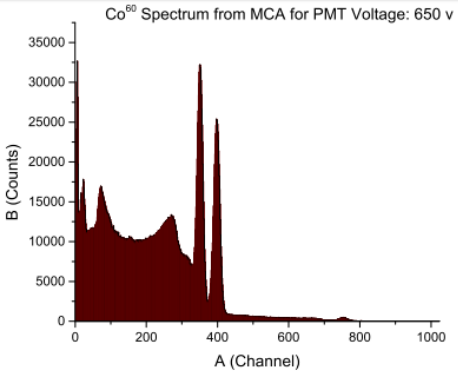
(b) Co^{60} spectrum for PMT voltage: 620 v



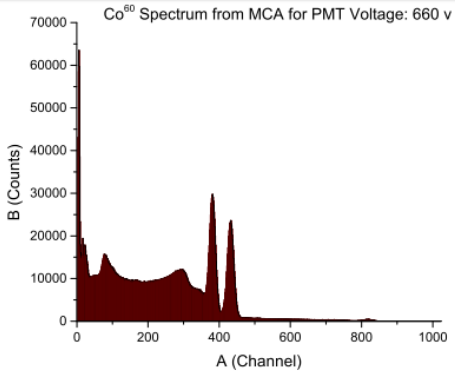
(c) Co^{60} spectrum for PMT voltage: 630 v



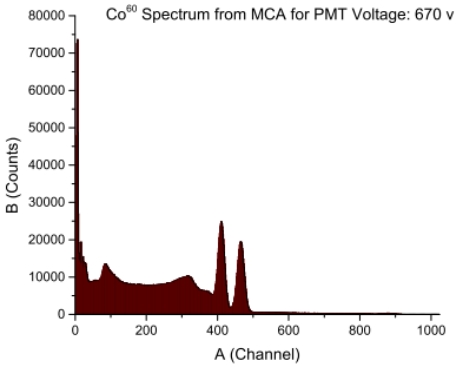
(d) Co^{60} spectrum for PMT voltage: 640 v



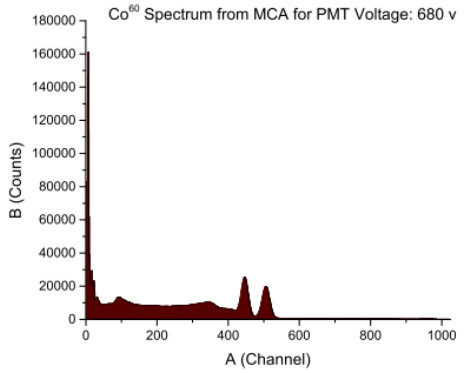
(e) Co^{60} spectrum for PMT voltage: 650 v



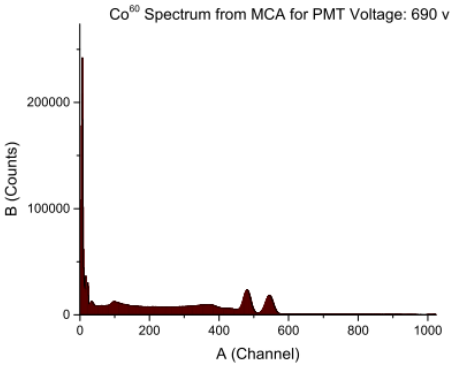
(f) Co^{60} spectrum for PMT voltage: 660 v



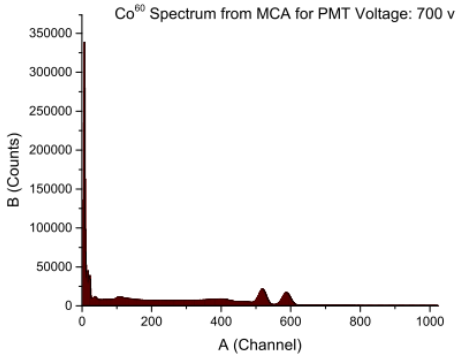
(g) Co^{60} spectrum for PMT voltage: 670 v



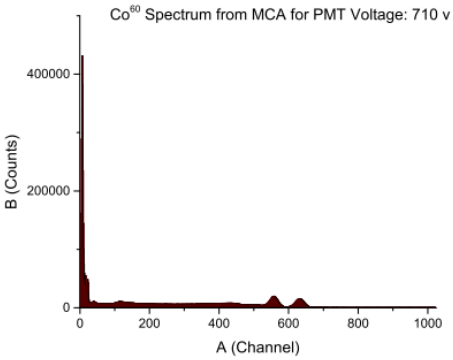
(h) Co^{60} spectrum for PMT voltage: 680 v



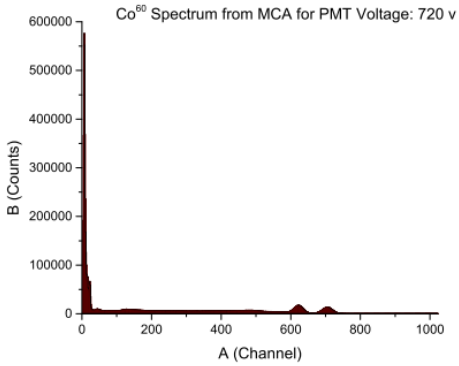
(i) Co^{60} spectrum for PMT voltage: 690 v



(j) Co^{60} spectrum for PMT voltage: 700 v



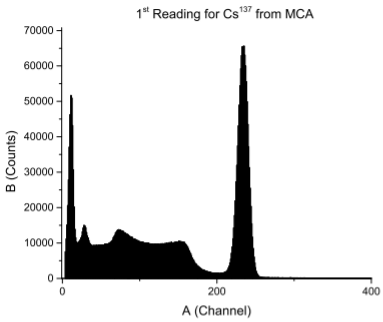
(k) Co^{60} spectrum for PMT voltage: 710 v



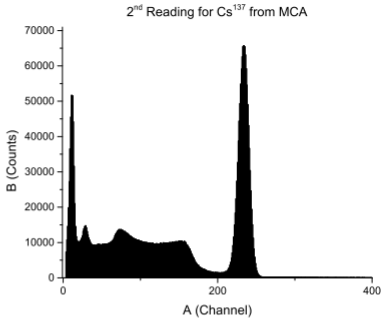
(l) Co^{60} spectrum for PMT voltage: 720 v

(k) Co^{60} spectrum for PMT voltage: 710 v

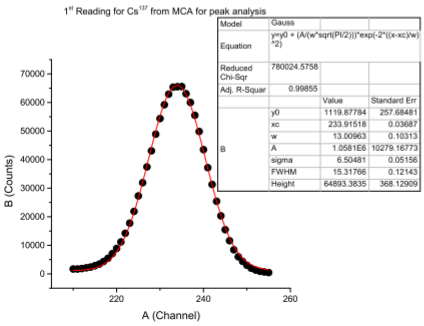
(l) Co^{60} spectrum for PMT voltage: 720 v



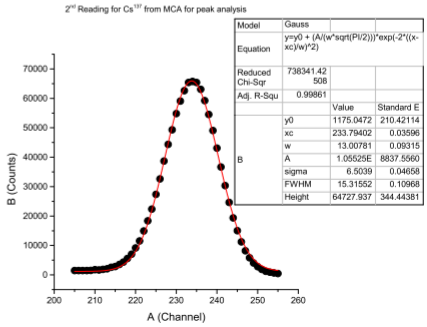
(m) 1st Reading from MCA for Cs^{137} source



(n) 2nd Reading from MCA for Cs^{137} source



(m) 1st Reading from MCA for Cs^{137} source



(n) 2nd Reading from MCA for Cs^{137} source

Figure 4: Gamma Energy Spectrum of Co^{60} and Cs^{137} from MCA

3.2.2 Calibration of MCA and finding photopeak energy of Cs^{137}

- **Calibration of MCA:** We will calibrate MCA at $670V$ PMT voltage. We discussed calibration of MCA in section 2.4. Value of energy E_A and E_B for Co^{60} photopeaks are known to be 1.17 MeV and 1.33 MeV respectively. We use peak finding tool in origin to find photopeack chanel number A and B. From figure-5 we see that value of A and B are 412 and 467 repectively. Hence the calibration constant α discuused in section 2.4 is

$$\alpha = \frac{E_A - E_B}{A - B} = \frac{1.33 - 1.17}{467 - 412} = 0.0029090909 MeV$$

- **Photopeak energy of Cs^{137}:** From figure 4(m) and 4(n) we find CS^{137} photopeak chanel number to be 234 (both data gives same result). Hence the corresonding energy is

$$E_{Cs} = \alpha \times 234 = 0.68072 Mev$$

3.2.3 Photopeak Efficiency of Co^{60} peaks

Photo peak efficiency is defined as

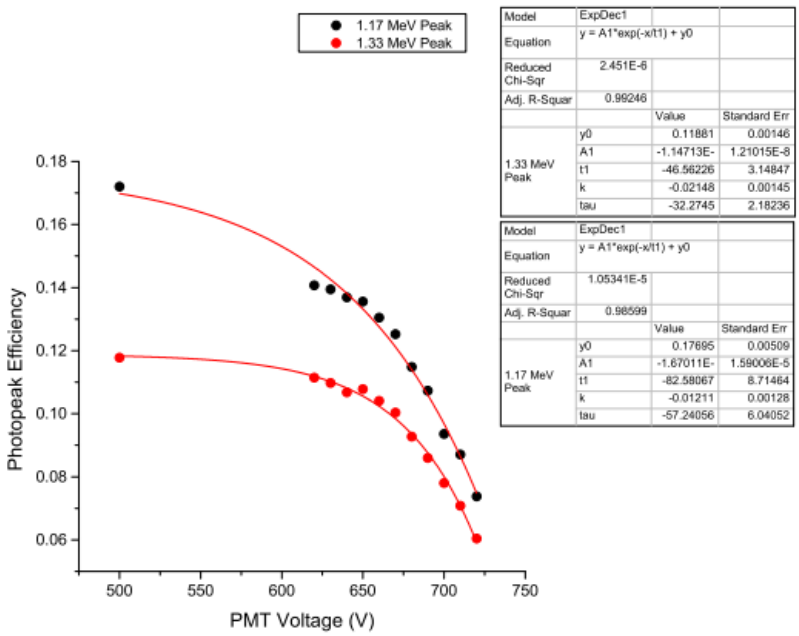
$$Photopeakefficiency = \frac{Area\ under\ the\ Peak}{Area\ under\ the\ Whole\ spectrum} \tag{4}$$

We integrate the resulting spectrum from MCA is in order to find out the area. The result obtained is presented in table-2.

PMT Voltage	**Area under the curve**			**Photopeak Efficiency**	
	Whole Spectrum	**1.17 MeV Peak**	**1.33 MeV Peak**	**1.17 MeV Peak**	**1.17 MeV Peak**
500	4524150	778507	532998	0.1720780699	0.1178117436
620	4914210	691507	547729	0.1407158017	0.1114581998
630	4952400	690819	543563	0.1394917616	0.1097574913
640	5055000	692103	539923	0.1369145401	0.1068096934
650	5228720	709018	563497	0.1356006824	0.107769588
660	5404180	705313	562321	0.1305124922	0.1040529738
670	4982830	623862	499929	0.1252023449	0.1003303344
680	5942860	682280	551216	0.1148066756	0.09275264772
690	6344850	680686	545637	0.1072816536	0.08599683208
700	6896630	645249	537899	0.0935600431	0.07799446976
710	7484110	651527	529736	0.0870547066	0.07078142892
720	8555230	631211	516869	0.0737807166	0.0604155586

Table 2: Data for photopeak Efficiency of Co^{60} peaks at different PMT Voltages

We plot efficiency values vs PMT voltage. The resulting graph, as in Figure-5, is fitted with exponential decay model (i.e. $y = y_0 + Ae^{xt}$), which is a steeply decreasing function. As it can be seen from figure-5 rate of decrement is more for 1.17 MeV Peak than 1.33 MeV Peak, hence difference between efficiency value of two peaks also decreases rapidly towards higher voltage.



Model	ExpDec1		
Equation	y = A1*exp(-x/t1) + y0		
Reduced Chi-Sqr	2.451E-6		
Adj. R-Squar	0.99246		
		Value	Standard Err
1.33 MeV Peak	y0	0.11881	0.00146
	A1	-1.14713E-	1.21015E-8
	t1	-46.56226	3.14847
	k	-0.02148	0.00145
	tau	-32.2745	2.18236

Model	ExpDec1		
Equation	y = A1*exp(-x/t1) + y0		
Reduced Chi-Sqr	1.05341E-5		
Adj. R-Squar	0.98599		
		Value	Standard Err
1.17 MeV Peak	y0	0.17695	0.00509
	A1	-1.67011E-	1.59006E-5
	t1	-82.58067	8.71464
	k	-0.01211	0.00128
	tau	-57.24056	6.04052

Figure 5: Gamma Energy Spectrum of Co^{60} and Cs^{137} from MCA

3.2.4 Variation of resolution with PMT voltage

We can determine resolutio (R) using the formula

$$R = \frac{\delta E}{E} \times 100\% \tag{5}$$

where,
δE = The full width of the peak at half of the maximum
E = The photopeak chanel number.
Note Energy and chanel number are related by a constant factor so $\frac{\delta CN}{CN} = \frac{\delta E}{E}$ Where CN denotes chanel number. We analyze the the part of the spectrum from MCA in the vicinity of photopeak chanel number by dual peak analysis tool from Origin software, We fitted the the baseline with parabolic function. We subtracted the baaseline function from the whole data and resulting data is fitted with sum of two independent Gaussian function. From the fitted guassian we can find chanel number corresponding to photopeack(E) and full width at half maxima. The result is presented in table-3.

PMT High Voltage (V)	ΔE_1	E1(CN)	Resolution (r1)	ΔE_2	E2(CN)	Resolution (r2)
500	5	79	0.06329113924	5	91	0.05494505495
620	13	260	0.05	14	295	0.04745762712
630	14	283	0.04946996466	15	321	0.04672897196
640	17	323	0.05263157895	18	364	0.04945054945
650	18	351	0.05128205128	19	398	0.04773869347
660	19	381	0.0498687664	21	432	0.04861111111
670	20	411	0.04866180049	22	466	0.04721030043
680	22	447	0.04921700224	25	506	0.04940711462
690	24	481	0.0498960499	27	545	0.0495412844
700	25	519	0.04816955684	29	588	0.04931972789
710	28	558	0.05017921147	30	632	0.04746835443
720	31	623	0.04975922953	34	705	0.04822695035

Table 3: Resolution of Co^{60} peaks at different PMT voltages

We plot the reolution vs PMT voltage. The resulting plot is given in Figure-6. Linear fit of the data shows that, resolution of 1.17 MeV peak is around 0.05917, whereas that of 1.33 MeV peak is around 0.0417 (intercept values; as both fits are with almost horizontal due to slope almost 0). Hence MCA instrument has better resolution for 1.17 MeV.

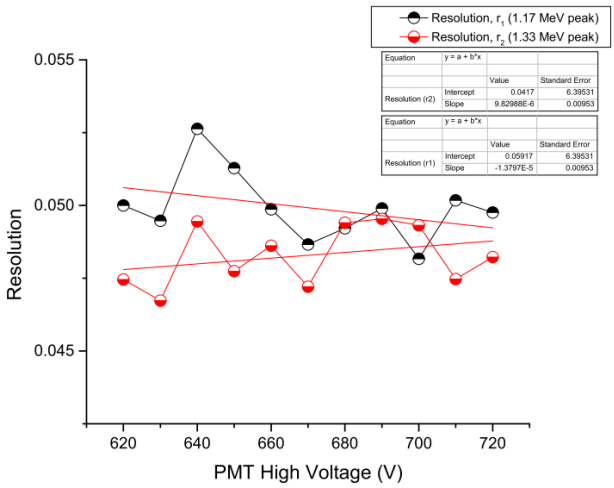


Figure 6: Plot of resolution of Co^{60} peaks at different PMT voltage

3.2.5 Variation of calibration constnat (α) with applied PMT voltage

Since photopeak energy E_1 and E_2 does not change with change of PMT voltage and as we observe in table-3 chanel number corresponding to E_1 and E_2 changes. This is possible only if calibration constant

changes with change of applied PMT voltage(V). Let $\alpha(v)$ denote the value of the cailbration constant at PMT voltage v. Then data E_1 and E_2 listed in table-3 can be expressed as

$$E_1(CN) = \frac{E_1}{\alpha(v)} = \frac{1.17}{\alpha(v)} \tag{6}$$

$$E_2(CN) = \frac{E_2}{\alpha(v)} = \frac{1.33}{\alpha(v)} \tag{7}$$

variation of $E_1(CN)$ and $E_2(CN)$ with PMT voltage v is shown in figure-7.

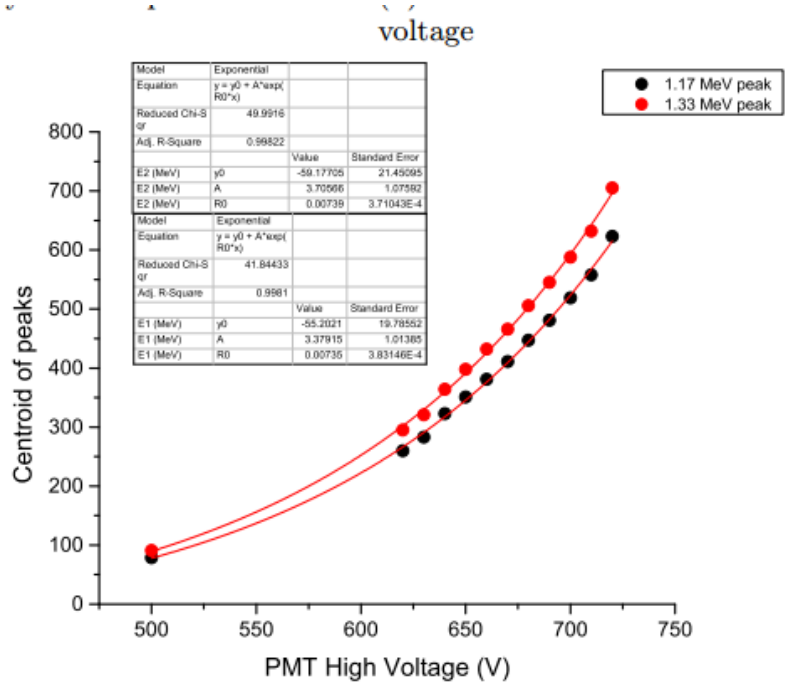


Figure 7: Position of centroid of Photo peaks for Co^{60}

3.2.6 Analysis of Co^{60} spectrum

We choose to analyse Co^{60} spectrum at PMT voltage= 670V. As it can been seen from figure-1(g) we observe a backscattering peak, compton edge peak and two photopeak. Co^{60} decays by beta decay to the stable isotope nickel-60. The activated nickel nucleus emits two gamma rays with energies of 1.17 MeV and 1.33 MeV. Two photopeaks corespond to these two gamma energy. We calculate compton edge energy for these two gamma rays using the formula $\frac{2E_\gamma^2}{2E_\gamma+mc^2}$ It comes out to be = 0.9671MeV for 1.1732 MeV gamma ray and 1.1220MeV, for 1.3325 MeV gamma ray. Corresponding chanel numbers are 328 and 380. We calculate backscattering energy using the formula $\frac{mc^2E_\gamma}{2E_\gamma+mc^2}$. It comes out to be 0.2061MeV for 1.1732 MeV gamma ray and = 0.2105MeV, for 1.3325 MeV gamma ray. Corresponding chanel numbers are 70 and 71. Chanel numbers are obtained simply by deviding corresponding energy with calibration constant α at 670V. Thus we expect two backscatter peak to to produce a pronounced peak as they nearly coincide. Since two compton edge differ in chanel number by 52 we expect to observe a distributed compton peak. This is observed in figure-1(g)a pronounced peak is observed at around chanel number 85 and a ditributed peak around chanel number 322.

As discuused in section 2.1.3 for energy $> 1.02MeV$ we expect pair production peaks.

1. For 1.17 MeV γ ray, Full Energy Peak is 1.1732 MeV, One-escape Peak is 0.6650 MeV ($E - m_e c^2$ =1.1732-0.5082 MeV), Two-escape Peak is 0.1567 MeV ($E - 2m_e c^2$ =1.1732-1.0165 MeV).

2. For 1.33 MeV ray, Full Energy Peak is 1.3325 MeV, One-escape Peak is 0.8243 MeV ($E - m_e c^2$ =1.3325-0.5082 MeV), Two-escape Peak is 0.3160 MeV ($E - 2m_e c^2$ =1.3325-1.0165 MeV)

Channels corresponding to energy 0.6650, 0.1567, 0.8243, 0.3160 MeV are 225, 53, 279, 107. Except channel 53, rest are within Compton Region. So only pair production peak that can be observed is peak near chanel number 53. We do see a peak in figure-1(g) near chanel number 53.

4 Discussion

Otimised value of the parametrs are to be used . For example if we set large gain from PMT it may lead to amplifi-cation of noise over actual data, leading to erroneous conclusion and if we set the gain too low it would not give the signal strong enogh to be observed. Gain value of 8 is optimised value. As the nuclear daecay is a probabilistic proccess each data is to be taken for long time. Data is taken over 600 seconds interval to ensure proper statistical interpretation, not instantaneous observation. After switching-on the measuring instruments and placing the source on top of Scintillator, we do not take ant data to ensure full ionization of the NaI:Tl crystal.

- Obtained value of Cs^{137} photopeak is 0.68 MeV and the literature value is 0.66164 MeV which is very close. From this we can say that our calibratiob is quite good.

- From figure-7 we observe that $E_1(CN)$ and $E_2(CN)$ increases with increase of PMT voltage hence using equation(6) and (7) we can say that calibration constant decreases with increase of PMT voltage. Equation (6) and (7) also ecplains why E_2(CN)>E_1(CN) it's because of praportinality constant 1.33 in case of E_2(CN) compared to 1.17 in case of E_1(CN).

- From figure-4(a) to 4(i) and also from total spectrum area in table-2 we see that number of counts increases with increase of PMT voltage. However nature of spectrum deviates fromthe theoretical one if PMT voltage is either too less or too high from 670V. SO MCA works as required best at 670V PMT voltage. This is why we choose 670V for calibration of MCA.

5 Conclusion

- We report energy of gamma ray emitted in radioactive emission of Cs^{137} to be 0.68 MeV.

- Optimum PMT voltage of given MCA instruement is 670V.